1. Two questions, seven tasks

Research question: Does relabeling improve utility, and does fitting the generator with Y help?

Method: Use verified completed CTGAN/TVAE runs, seeds 42/43 and RF/XGB/DNN. Keep synthetic-only and mix separate.

Finding: Classification gains favor relabeling but do not pass Holm across the two questions. Average has_y effects are small. Regression depends on the task and metric.

Design choice	Definition
Classification tasks	Adult, Census KDD, Covertype, MNIST12, MNIST28. Outcome: macro F1 for all five.
Regression tasks	News and California Housing. Outcomes: saved R2, NMAE and absolute-error D2, separately.
Relabeled	Average both has_y settings, both generators and RF/XGB/DNN; keep generated X and predict Y.
Generated-target comparator	Fit each generator to (X,Y), retaining its generated Y. Match dataset, seed and generator.
has_y=1 versus has_y=0	Compare generator fitted to (X,Y) versus fitted to X only. Both arms receive predicted targets.
Training regimes	Synthetic-only: 100,000 synthetic rows. Mix: all real training rows plus 100,000 synthetic rows.
Coverage	406 selected completed records: 290 classification and 116 regression, including 14 shared real-only references.

```
C_relabel[d] = mean(relabel[d]) - mean(generated[d])
```

```
C_has_y[d] = mean(relabel[d, has_y=1]) - mean(relabel[d, has_y=0])
```

Means give equal weight to seeds, generators and labelers. Classification inference then gives each of the five datasets equal weight. Each generated baseline is a shared comparator, not six independent observations.

The two questions were selected after earlier analyses of these results. This redo is exploratory; it is not a newly preregistered confirmatory analysis. It excludes Credit, Intrusion, NB/PCA-GMM, simulated benchmarks and active TabDDPM runs.

2. Classification: gains and evidence

Research question: Are either of the two average macro-F1 contrasts supported across tasks?

Method: Two-sided paired t-tests on five dataset differences, df=4. Holm2 covers both questions within a regime; Holm4 covers all four tests across both regimes.

Finding: Relabeling gains 11.63 points for synthetic-only and 6.77 for mix; all five task gains are positive. Neither relabeling nor has_y passes the Holm2 threshold of .05.

Training	Contrast	Gain (points)	Raw p	Holm2 p	Holm4 p
Synthetic-only	Relabeled - generated	+11.63	0.0350	0.0700	0.1399
Synthetic-only	has_y=1 - has_y=0	+0.36	0.4575	0.4575	0.9149
Mix	Relabeled - generated	+6.77	0.0566	0.1132	0.1698
Mix	has_y=1 - has_y=0	+0.14	0.7182	0.7182	0.9149

Task	Synthetic relabel	Mix relabel	Synthetic has_y	Mix has_y
Adult	+3.53	+1.71	-0.69	-0.75
Census KDD	+3.28	+4.80	+0.15	-0.06
Covertype	+11.76	+13.63	+1.87	+1.44
MNIST12	+21.58	+12.03	+0.62	+0.07
MNIST28	+18.02	+1.70	-0.17	-0.01

Every change in the second table is in macro-F1 percentage points; positive favors relabeling or has_y=1. The has_y contrast is positive on three of five synthetic-only tasks and two of five mix tasks.

Synthetic-only mean F1: relabeled .7855 versus generated .6692. Mix: .8166 versus .7488. The corresponding relative improvements are 17.38% and 9.04%; relative percentages and percentage points are different quantities.

Raw synthetic relabel p=.0350 is below .05; its Holm2 p=.0700 is above .05. Holm changes the inferential threshold, not the gain. No detected has_y effect does not establish equivalence.

3. Regression: News and Housing

Research question: Do the same constructions improve regression utility?

Method: Compare native-unit means across seeds, generators and RF/XGB/DNN separately for each task. These baseline contrasts are descriptive; no seven-task pooled score is computed.

Finding: Housing relabeling improves R2, NMAE and D2. News relabeling improves R2 but worsens NMAE and D2. has_y effects are smaller and have mixed directions.

Synthetic-only

Task / metric	Generated	Relabeled	has_y=0	has_y=1	Relabel change	has_y change
News / R2	-0.0235	0.0011	0.0027	-0.0004	+0.0246	-0.0031
News / NMAE	0.2546	0.2592	0.2597	0.2587	+0.0046	-0.0010
News / D2	0.0099	-0.0079	-0.0098	-0.0060	-0.0178	+0.0038
Housing / R2	0.6730	0.7938	0.7946	0.7930	+0.1208	-0.0015
Housing / NMAE	0.4146	0.3036	0.3049	0.3022	-0.1111	-0.0027
Housing / D2	0.4595	0.6043	0.6025	0.6061	+0.1448	+0.0036

Mix

Task / metric	Generated	Relabeled	has_y=0	has_y=1	Relabel change	has_y change
News / R2	-0.0196	-0.0003	0.0002	-0.0008	+0.0192	-0.0010
News / NMAE	0.2494	0.2548	0.2551	0.2544	+0.0054	-0.0006
News / D2	0.0300	0.0092	0.0079	0.0104	-0.0208	+0.0025
Housing / R2	0.6676	0.7888	0.7905	0.7872	+0.1212	-0.0033
Housing / NMAE	0.4137	0.3077	0.3103	0.3052	-0.1060	-0.0051
Housing / D2	0.4607	0.5989	0.5956	0.6022	+0.1382	+0.0066

Higher R2 and D2 is better; lower NMAE is better. The change columns retain native signs: negative NMAE change is improvement. R2 uses squared error against a test-mean baseline; D2 uses absolute error against a test-median baseline.

NMAE = saved MAE / saved SD_test(Y), with population SD (ddof=0). Saved record normalization and verified D2 sidecars are read directly; denominators are not recomputed. News log-target runs are scored in original units; Housing uses no-Faker runs.

Only two fixed regression tasks and two seeds are available. Descriptive improvement on Housing does not establish a population effect on new regression tasks. Testing multiple regression outcomes would also require an explicit family if used for joint significance claims.

4. Exploratory factor checks

Research question: What do factorial models add beyond the two research contrasts?

Method: Refit categorical Type II ANOVAs using sum-to-zero coding. Generated targets stay outside the relabeled factorial grid. All p-values on this page are nominal and exploratory.

Finding: Labeler choices differ in the classification model and treatment effects vary strongly by task. These tests do not establish improvement over generated targets.

```
macro_F1 ~ dataset + generator * has_y * labeler * seed
```

Classification: 120 relabeled scores per regime, model rank 28 and residual df 92. The ANOVA assumes independent, homoskedastic residuals; shared heldout rows and task-specific effects limit its interpretation.

Classification term	Synthetic nominal p	Mix nominal p
dataset	1.17e-49	5.36e-52
generator	0.7746	0.9987
has_y	0.6671	0.8459
labeler	1.03e-05	7.85e-06
seed	0.8926	0.8584

Supporting nested test	Synthetic nominal p	Mix nominal p
seed interactions	0.9998	0.9999
dataset interactions	1.86e-09	2.93e-32

```
R2 or D2 ~ dataset + generator * has_y * labeler * seed
```

Regression pooled checks: 48 relabeled scores per outcome/regime, rank 25, residual df 23. The two tasks remain fixed study tasks; pooled coefficients are not evidence about a population of regression tasks.

Training / outcome	Generator p	has_y p	Labeler p	Seed p
Synthetic-only / R2	0.2947	0.7332	0.0879	0.3036
Synthetic-only / D2	0.0164	0.4608	0.0475	0.4740
Mix / R2	0.0423	0.5589	0.0499	0.7202
Mix / D2	0.0503	0.2605	0.0005	0.5071

Complete ANOVA tables and within-task regression models are supplied as CSVs. Within each regression task: outcome ~ seed + generator * has_y * labeler, rank 13 and residual df 11. A full four-factor within-task model is saturated and has no residual degrees of freedom.

Nominal ANOVA p-values support exploration, not additional adjusted headline claims. A significant labeler term means differences among RF/XGB/DNN, not superiority to generated targets.

5. Uncertainty and exact sensitivity checks

Research question: How precise are the gains, and do small-sample checks agree?

Method: Show pointwise 95% paired-t intervals and Bonferroni simultaneous intervals for both questions within each regime. Enumerate all 32 sign patterns; compute exact Wilcoxon signed-rank p-values.

Finding: Simultaneous intervals for all four effects contain zero. Exact two-sided relabeling p=.0625 in each regime, before multiplicity adjustment.

Training / contrast	Pointwise 95% CI	Simultaneous 95% CI (2)
Synthetic-only / Relabeled - generated	[+1.33, +21.93]	[-1.33, +24.60]
Synthetic-only / has_y=1 - has_y=0	[-0.85, +1.56]	[-1.16, +1.88]
Mix / Relabeled - generated	[-0.31, +13.85]	[-2.14, +15.68]
Mix / has_y=1 - has_y=0	[-0.85, +1.13]	[-1.11, +1.38]

Intervals are in macro-F1 percentage points. Simultaneous intervals use t_(1-.05/(2*2),4); they are Bonferroni intervals, not an inversion of Holm. The package also includes simultaneous intervals for all four tests.

Training / contrast	Exact sign-flip p	Exact Wilcoxon p	Sign-flip Holm2
Synthetic-only / Relabeled - generated	0.0625	0.0625	0.1250
Synthetic-only / has_y=1 - has_y=0	0.6250	0.8125	0.6250
Mix / Relabeled - generated	0.0625	0.0625	0.1250
Mix / has_y=1 - has_y=0	0.7500	1.0000	0.7500

With five nonzero task effects, the minimum attainable two-sided exact p is 2/32=.0625. The prior normal-approximation Wilcoxon value for the all-positive relabeling effects was .0431; the exact value is .0625. Neither exact check rejects at .05.

The paired-t test assumes independent task-level differences with a suitable normal model; exact sign-flip and signed-rank checks assume independent differences symmetric about zero under the null. The task methods were not randomly assigned, so this is a symmetry-based sensitivity analysis, not a design-based randomization test.

MNIST12 and MNIST28 share a task family and prepared split, and MNIST12 TVAE tables retain documented holdout-feature collisions. Both facts weaken broad task-generalization claims. Additional experiment seeds do not supply additional independent datasets.

Two observed seeds share heldout partitions; the reported intervals do not quantify uncertainty from resampling new test rows. The results show observed gains with limited evidence about generalization, not evidence that the two constructions are equivalent.

6. Is Holm required for contrasts?

Research question: Does using a contrast automatically require Holm correction?

Method: Distinguish the individual estimand from the family of claims. Use the two-question family specified for this redo; also report the four-test family when drawing conclusions across both training regimes.

Finding: No. A contrast is a way of defining a comparison; multiplicity comes from making several inferential claims. Holm is a defensible choice for the two related questions, not the only valid procedure.

Situation	Implication
One prespecified primary hypothesis	Ordinary p-value and interval can be used for that single claim; no additional contrast adjustment is needed.
Several related claims	Define the family before interpreting significance. Holm controls the chance of at least one false rejection within that family, under valid marginal tests.
Prespecified or orthogonal contrasts	Planning improves focus but does not automatically give familywise error control. Correlated contrasts do not invalidate Holm.
Your two questions in one regime	Holm2. The smaller p must first be <=.025. Synthetic relabel raw p=.0350 misses this cutoff; mix raw p=.0566 also misses it.
Claims across both regimes	Holm4 covers the four reported classification tests. Separate model fits and shared data do not by themselves justify separate testing families.
Descriptive or exploratory results	Unadjusted p-values may be shown if labeled clearly; they should not be presented as familywise-controlled discoveries.
Current results already examined	Restricting to two questions now is transparent and useful, but cannot retroactively create prespecification. Retain two-sided tests in this redo.

The previous report tested four construction contrasts per regime. This report tests only relabeled versus generated and has_y=1 versus 0, as requested. It does not add separate full-relabel-versus-generated or X-only-relabel-versus-generated significance claims.

Holm2 and Holm4 are family-specific results, not interchangeable thresholds to choose after viewing p-values. If the scientific claim concerns either question in either regime, use Holm4. A joint conjunctive claim requiring all component tests to succeed is a different hypothesis and would need its own stated decision rule.

Provenance and checks: October 10, 2026, 11:35 PM Chicago. All 522 record/sidecar hashes matched; 3,770 frozen metric values matched source records. Eighteen models were refitted remotely with one numerical thread; all 202 ANOVA F statistics were independently reproduced by restricted least squares. No experiment, source record, split or research setting was changed.

References: R stats p.adjust documentation (stat.ethz.ch/R-manual/R-devel/library/stats/html/p.adjust.html); NIST, Bonferroni method (itl.nist.gov/div898/handbook/prc/section4/prc473.htm); SciPy, wilcoxon documentation (docs.scipy.org/doc/scipy/reference/generated/scipy.stats.wilcoxon.html).